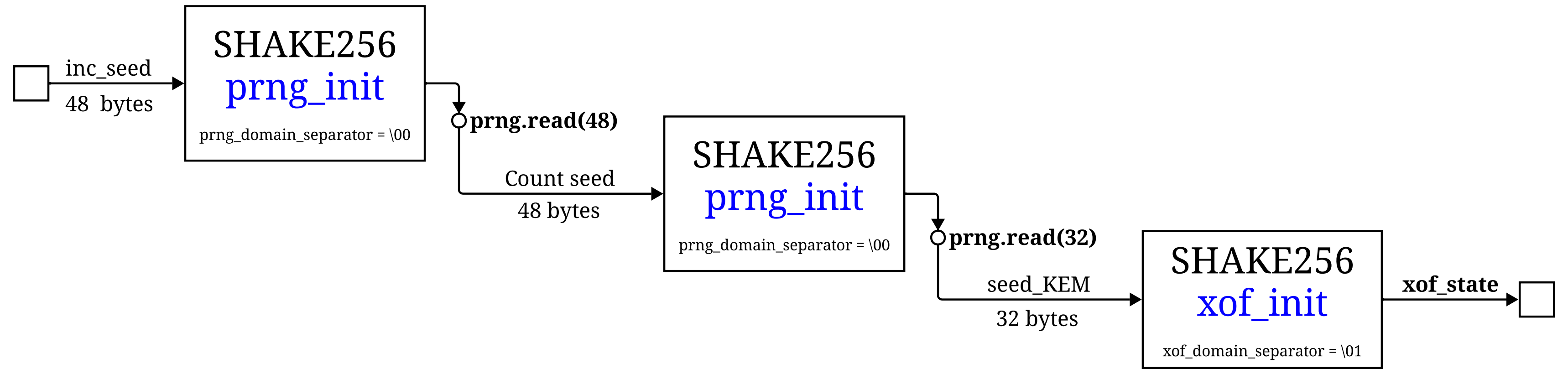
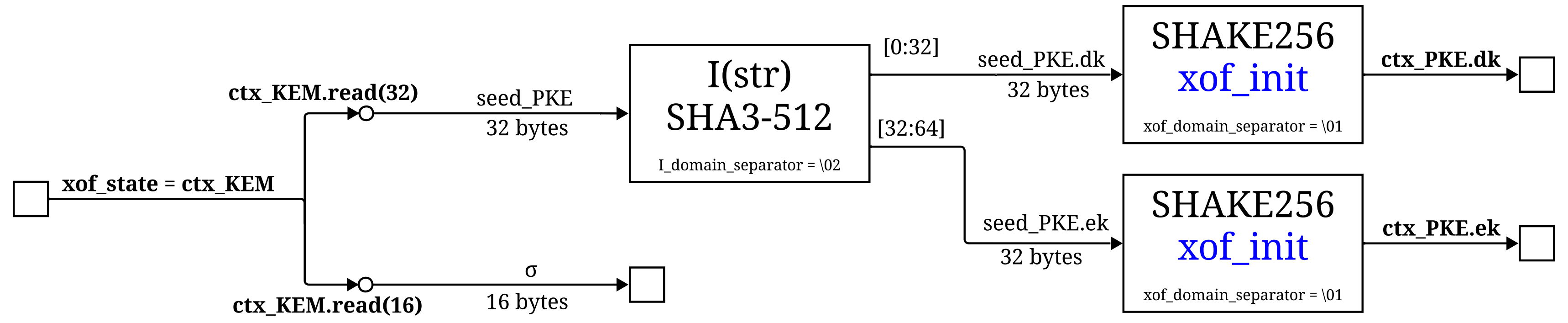
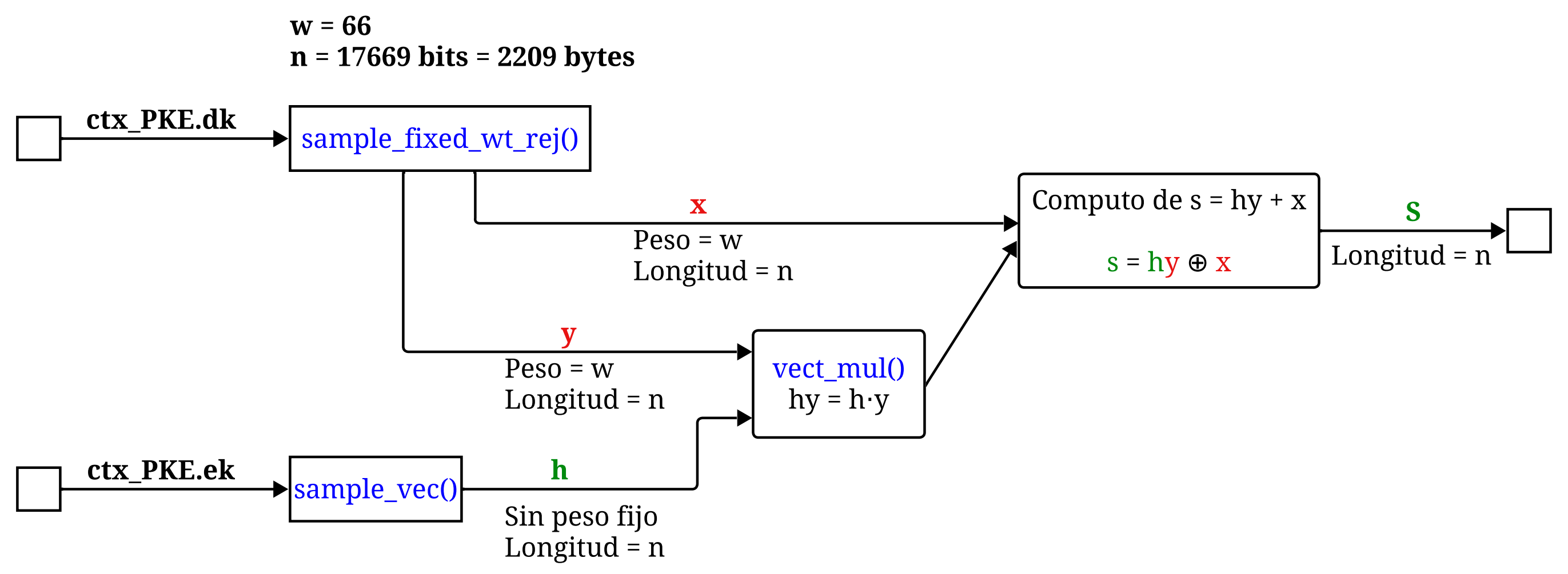
$$\mathbf{v} - \mathbf{u}\mathbf{y} = \mathbf{m}\mathbf{G} + \mathbf{s}\mathbf{r}_2 + \mathbf{e} - (\mathbf{r}_1 + \mathbf{h}\mathbf{r}_2)\mathbf{y} = \mathbf{m}\mathbf{G} + (\mathbf{x} + \mathbf{h}\mathbf{y})\mathbf{r}_2 + \mathbf{e} - \mathbf{r}_1\mathbf{y} + \mathbf{h}\mathbf{r}_2\mathbf{y}.$$

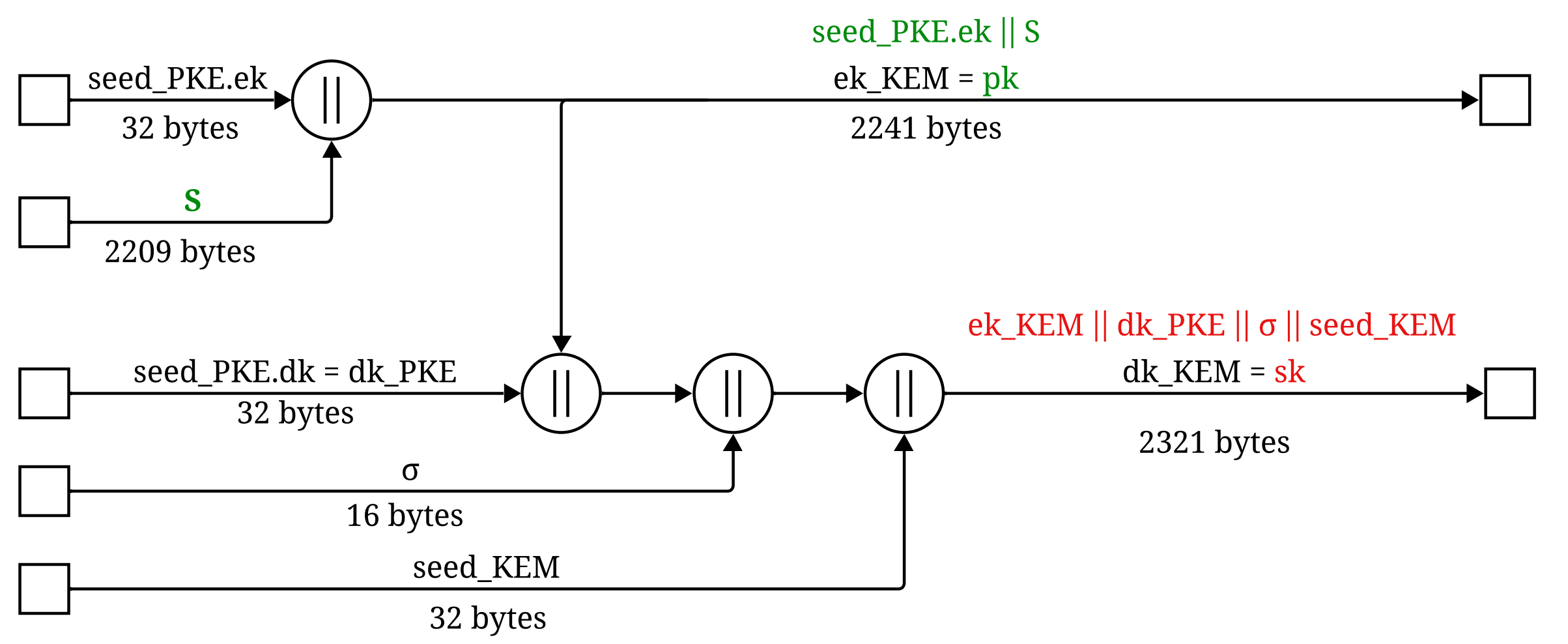
The remaining term $\mathbf{m}\mathbf{G} + \mathbf{x}\mathbf{r}_2 + \mathbf{e} - \mathbf{r}_1\mathbf{y}$ contains only a codeword plus sparse-sparse multiplicative terms.

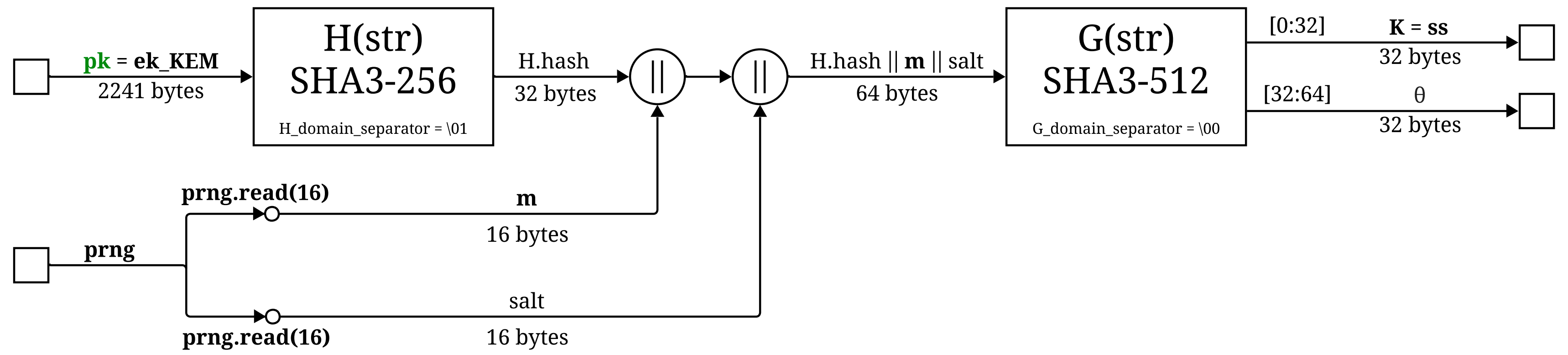
In other words, we have $\mathbf{v} - \mathbf{u}\mathbf{y} = \mathbf{m}\mathbf{G} + \mathbf{e}'$, and we just need to bound the weight of $\mathbf{e}'$.

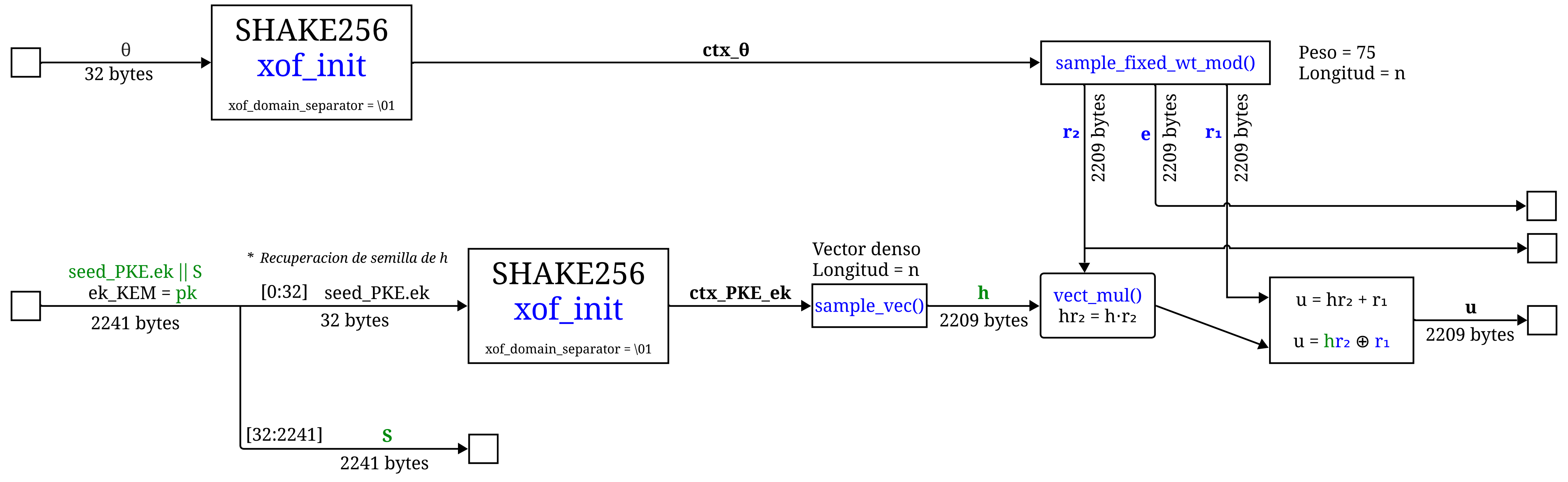


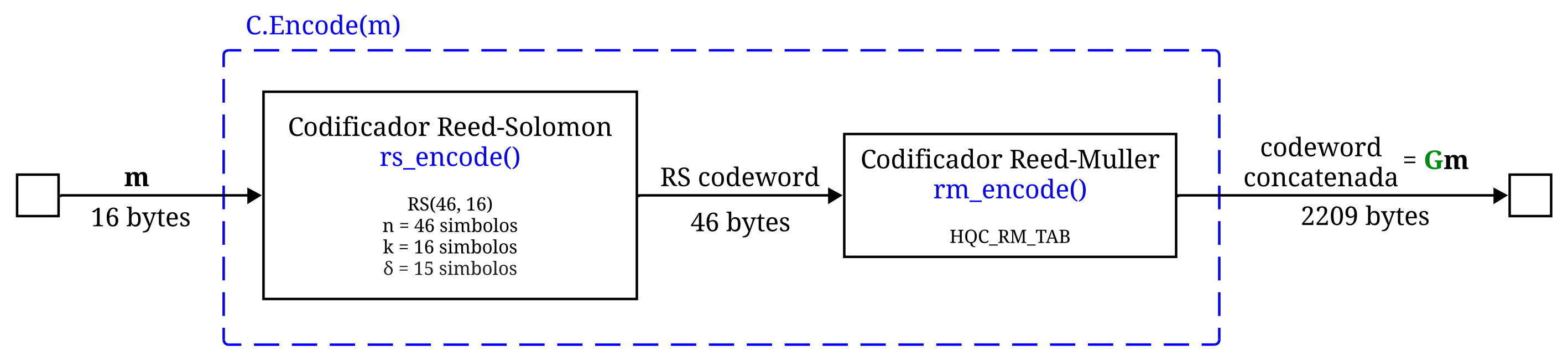


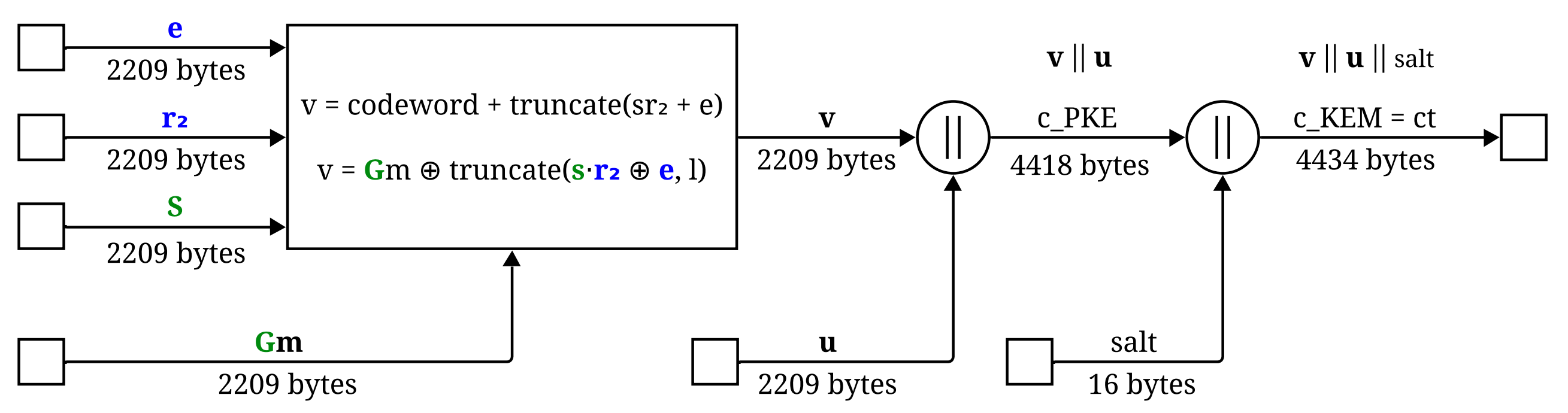


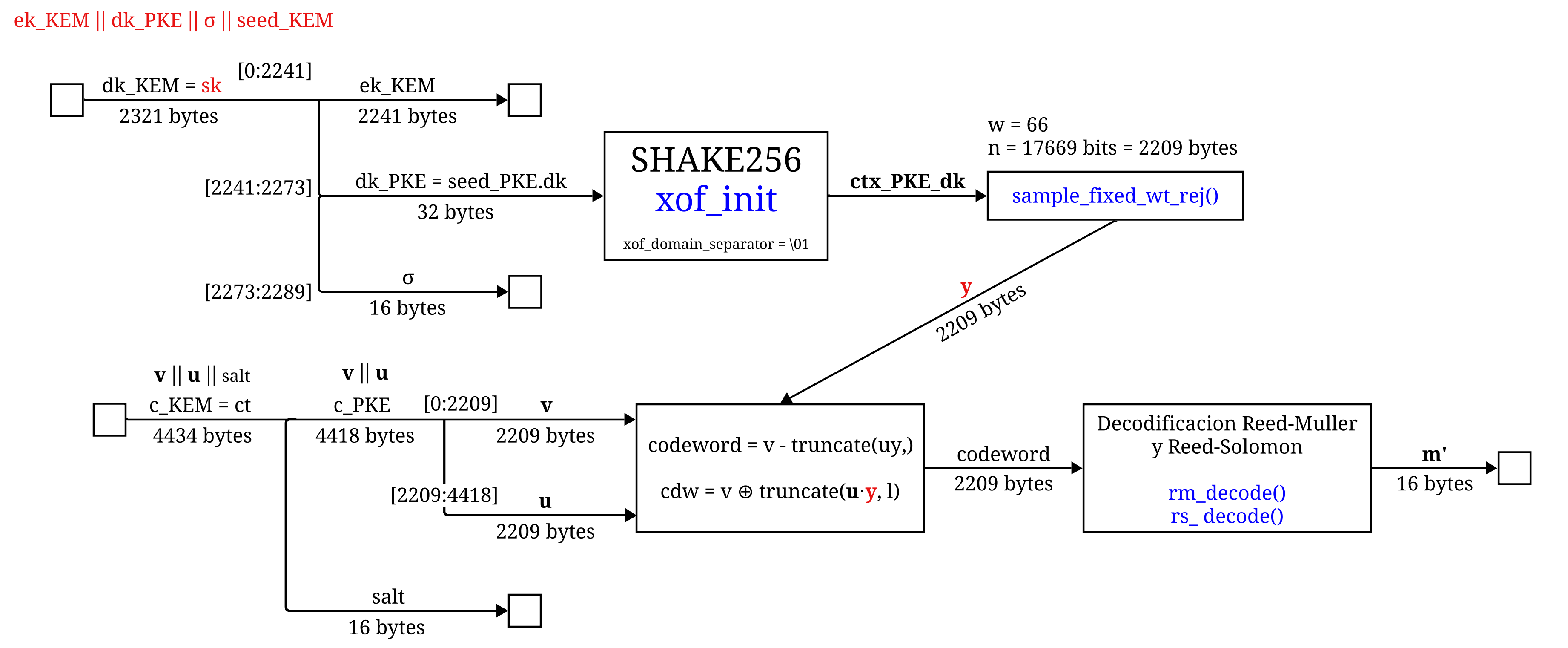


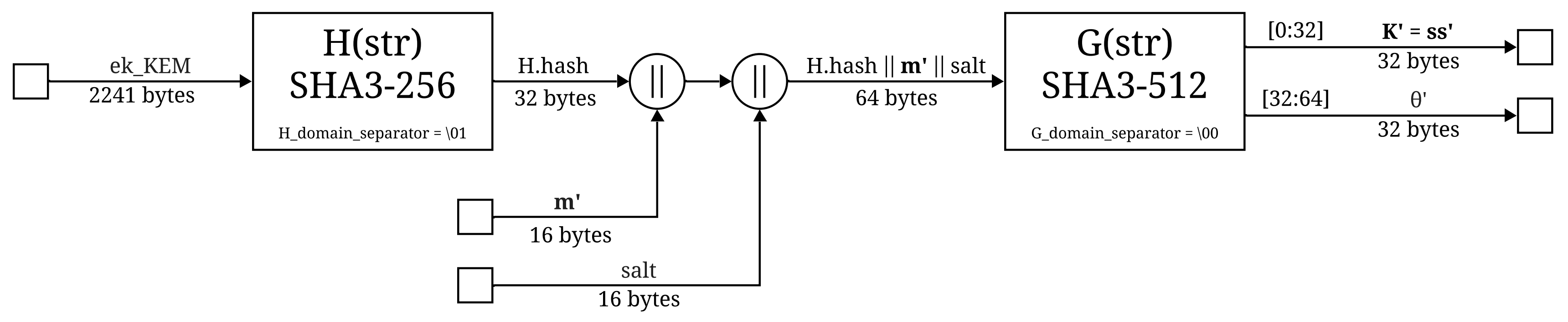


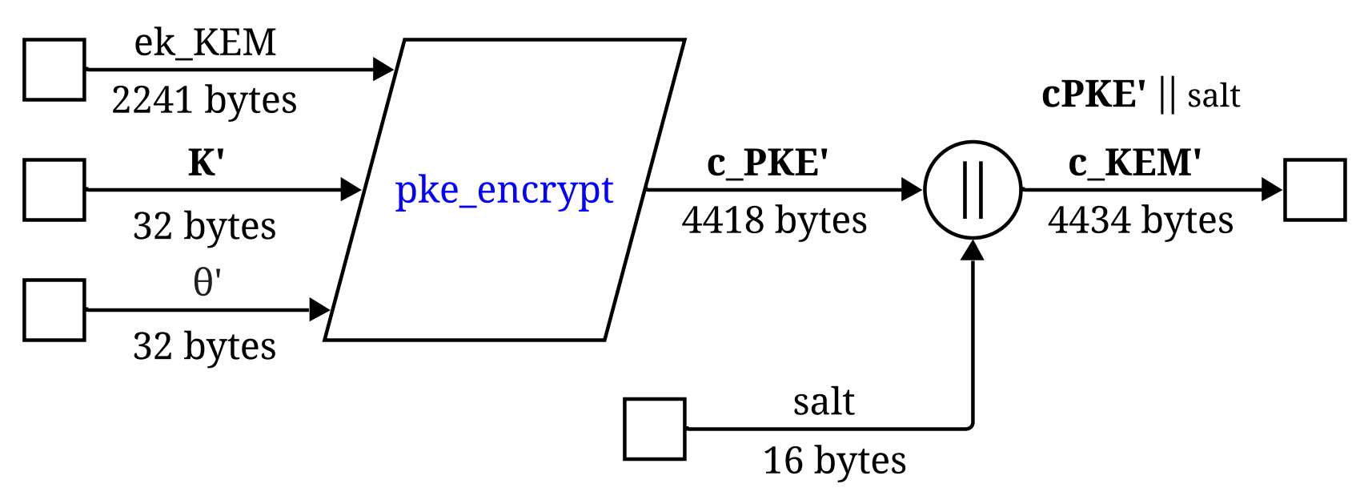


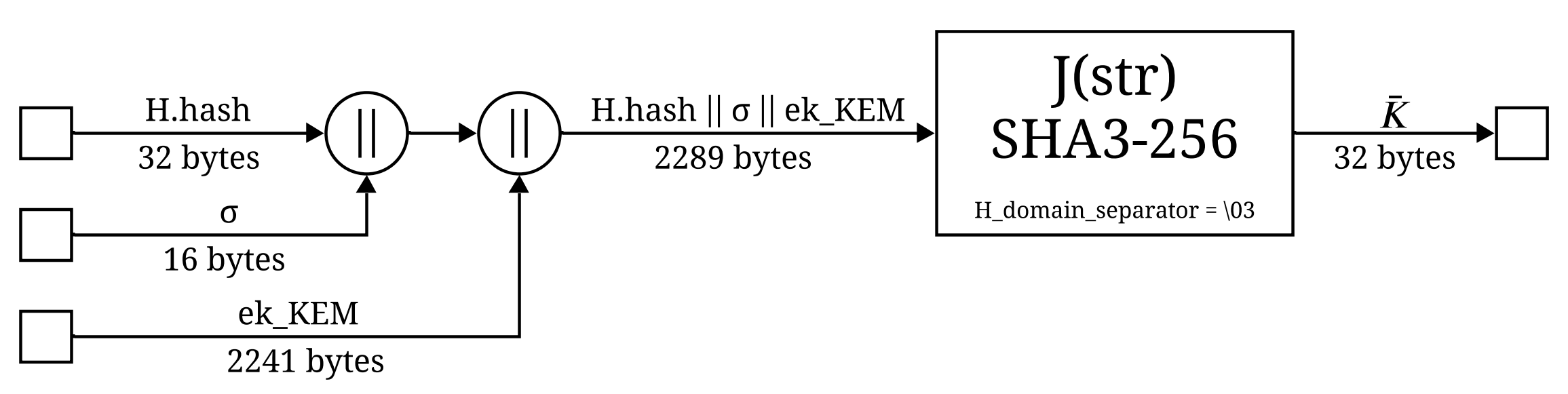












Verificación

m' = ⊥ ∨ c'_KEM ≠ c_KEM

K' ←

$$K' = \begin{cases} \bar{K}, & \text{si } m' = \perp \text{ o } c'_{\text{KEM}} \neq c_{\text{KEM}}, \\ K', & \text{en caso correcto,} \end{cases}$$